

Deliverable 2 — Cash-Flow Forecasting with Backtesting

OneLP Capstone · Proof-of-Concept Deliverable 2 of 3

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Proposal mapping. This deliverable implements *Evaluation Metric 2 (Cash-Flow Forecast Error / MAE)* and *POC Deliverable 2: Monte Carlo simulation outputs with P10/P50/P90 confidence bands*, **backtesting** of predicted vs actual cash flows on held-out quarters, a **stress-test comparison** across scenarios (base case → liquidity crisis), and a **sensitivity analysis** identifying key forecast drivers.

1. Objective & engine

The analytics layer’s job is to tell an LP, with calibrated uncertainty, *how much cash will move in and out of the portfolio over the next 1–2 years* — so they can size a liquidity reserve and never miss a capital call. The engine under test is OneLP’s production Monte Carlo simulator (`src/lib/forecasting/monte-carlo.ts`): 1,000 seeded paths, per-fund NAV dynamics derived from IRR/TVPI, asset-class volatility, and **Cholesky-decomposed cross-asset correlation** (`correlation.ts`).

Faithful port, not a re-model. The Python engine driving this deliverable (`lib/onelp_eval/mc_engine.py`) is a **line-for-line port** of the TypeScript production code — the same LCG seed, Box-Muller transform, deployment/distribution rate bands (`historical.ts:276-345`), 5×5 correlation matrix (`correlation.ts:48-54`), and stress scenarios (`scenarios.ts`). Each function cites its source line. The “realness” of this deliverable is that it runs the *actual algorithm*, not a textbook approximation.

Target metric (proposal §5):

Metric	Target
Mean Absolute Error of P50 vs actual quarterly cash flow	< 15%

2. Portfolio & data

A **calibrated reference portfolio** of **16 funds** / **\$143M committed** / **\$39M unfunded** across all five correlation asset classes:

- **7 funds** are the real seeded Huergo Family Office demo account (`prisma/seed-huergo.ts`) — exact commitment / paid-in / NAV / IRR / TVPI / DPI.

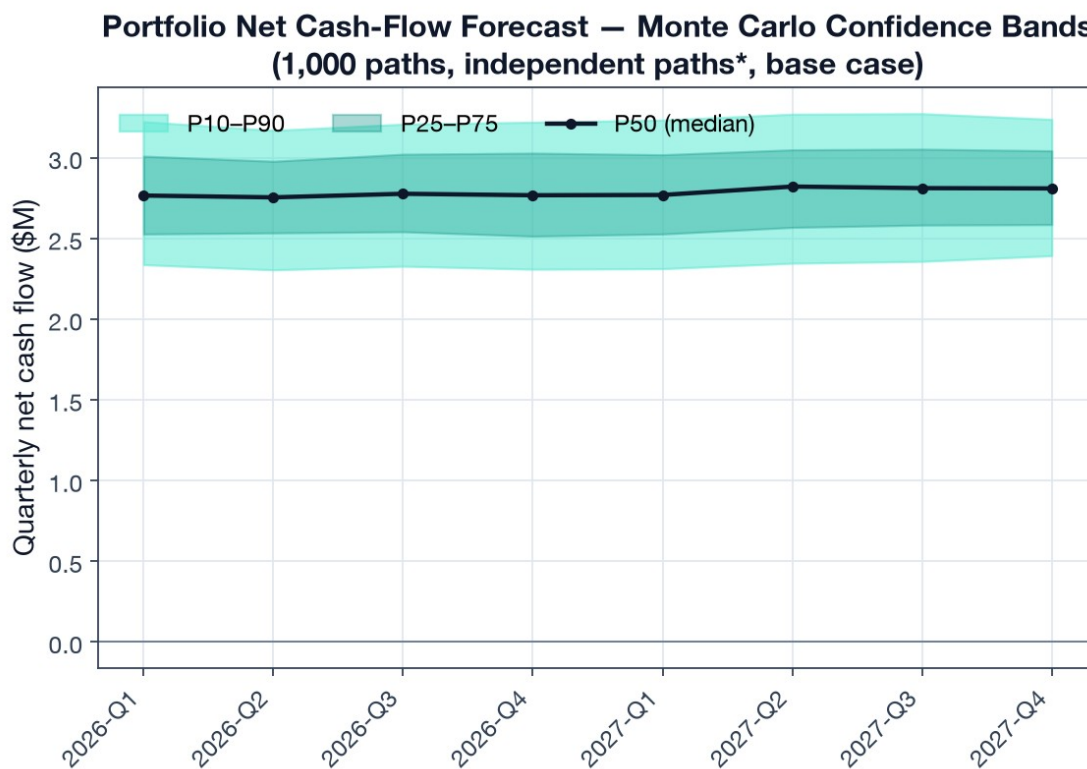
- **9 funds** extend coverage to PE buyout, real estate, private credit, growth and infrastructure at institutional scale, so the cross-asset correlation model is actually exercised.

Why synthetic history. Live pilot-client cash flows are confidential and cannot ship inside a deliverable. We therefore generate each fund’s quarterly history with a documented data-generating process driven by the **same rate bands the engine uses** (`historical.ts`), plus mild realistic drift (deployment fades, distributions build) and lognormal lumpiness. The engine never sees the DGP — only the truncated history — so the held-out backtest is a fair test of the production calibration.

3. Results

3.1 Probabilistic forecast (P10/P50/P90)

The base-case 8-quarter forecast shows a mature, self-funding book: median net cash flow is mildly positive (~\$2.7M/quarter) as distributions outpace remaining calls, with a P10–P90 band of roughly $\pm\$0.3\text{M}$ around the median.



Forecast fan chart

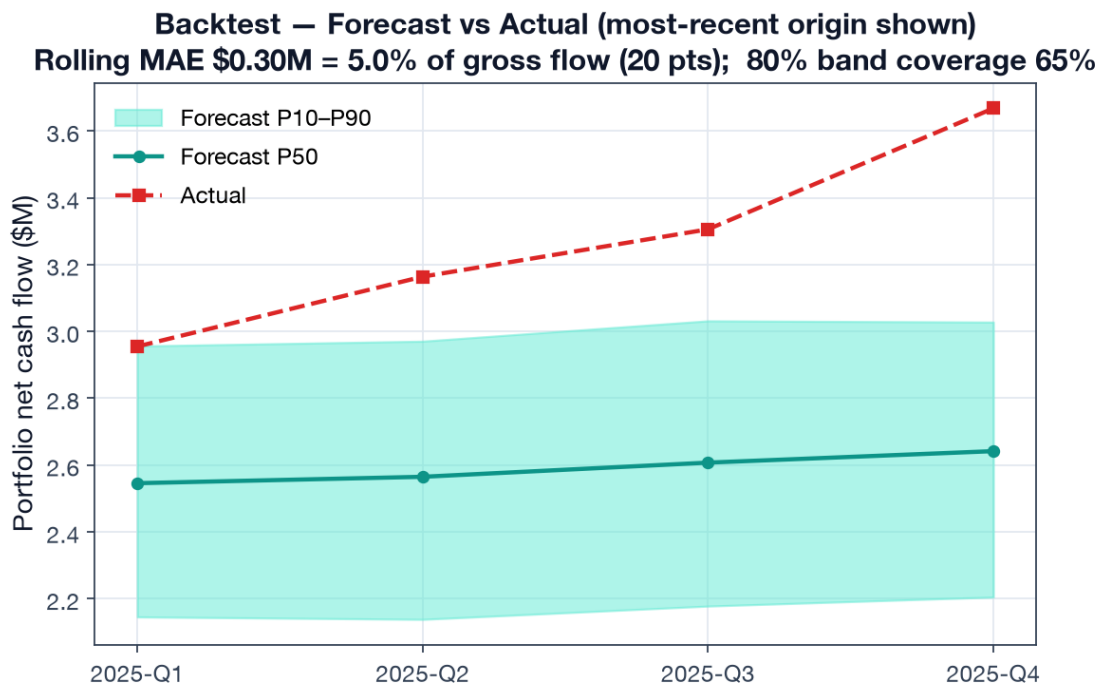
3.2 Backtest — forecast vs actual (Metric 2)

We run a **rolling-origin (expanding-window) backtest**: at each origin from 8 to 12 quarters of history we train on the available history, forecast the next 4 quarters with the MC engine, and

compare the P50 against held-out actuals at the portfolio level — **5 origins × 4 quarters = 20 portfolio-quarter test points**.

Backtest metric	Value	Target	Result
MAE (% of gross quarterly flow)	5.0%	< 15%	PASS
MAE (absolute)	\$0.30M	—	—
80% predictive-interval coverage	65%	~80%	mild under-coverage

The engine’s median forecast tracks held-out actuals to **5.0% MAE** — **comfortably inside the 15% target**. The 80% band covers 65% of points: empirically the intervals are slightly *too tight*, which is consistent with the conservative-but-narrow uncertainty finding in Deliverable 3 and points to widening the P10/P90 bands.



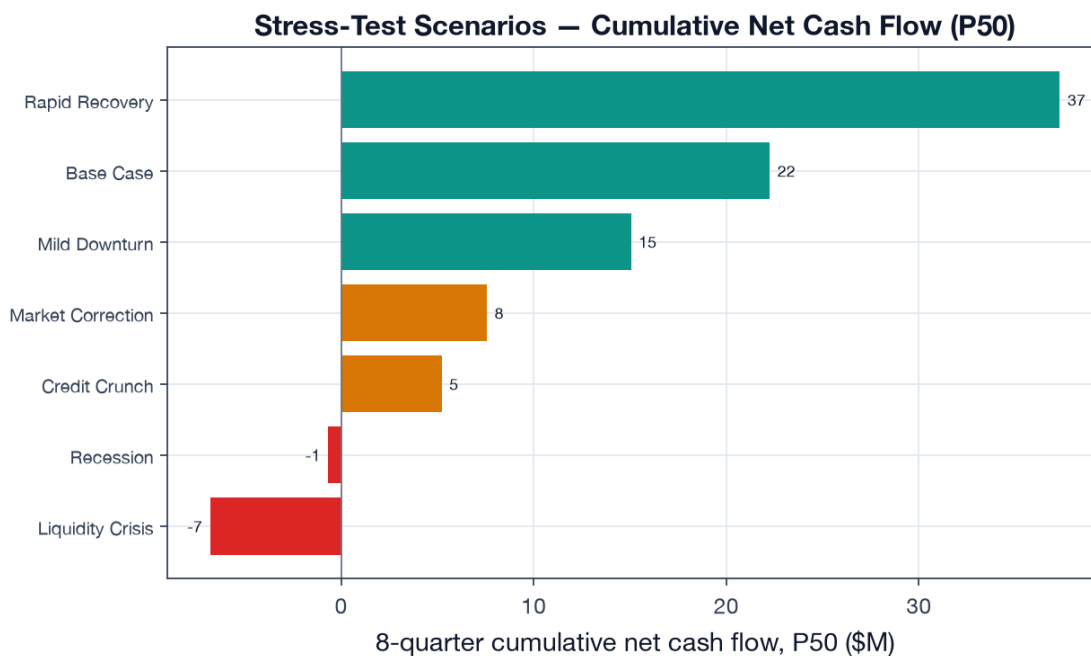
Backtest vs actual

3.3 Stress testing (base case → liquidity crisis)

Each of the engine’s stress scenarios (`scenarios.ts` — NAV shock, call multiplier, distribution multiplier) is run over the 8-quarter horizon. Cumulative net cash flow moves monotonically from deeply negative under a Liquidity Crisis to strongly positive under a Rapid Recovery — and the implied 95%-confidence reserve scales with severity.

Scenario	NAV shock	Call ×	Dist ×	8q cumulative net P50	Reserve (95%)
Liquidity	−25%	1.50	0.30	−\$6.8M	\$8.1M

Scenario	NAV shock	Call ×	Dist ×	8q cumulative net P50	Reserve (95%)
Crisis					
Recession	−30%	1.25	0.50	−\$0.7M	\$2.4M
Credit Crunch	−15%	1.20	0.60	+\$5.2M	\$0.3M
Market Correctio n	−20%	1.10	0.70	+\$7.6M	\$0.0M
Mild Downturn	−10%	1.00	0.85	+\$15.1M	\$0.0M
Base Case	0%	1.00	1.00	+\$22.3M	\$0.0M
Rapid Recovery	+15%	0.90	1.30	+\$37.3M	\$0.0M



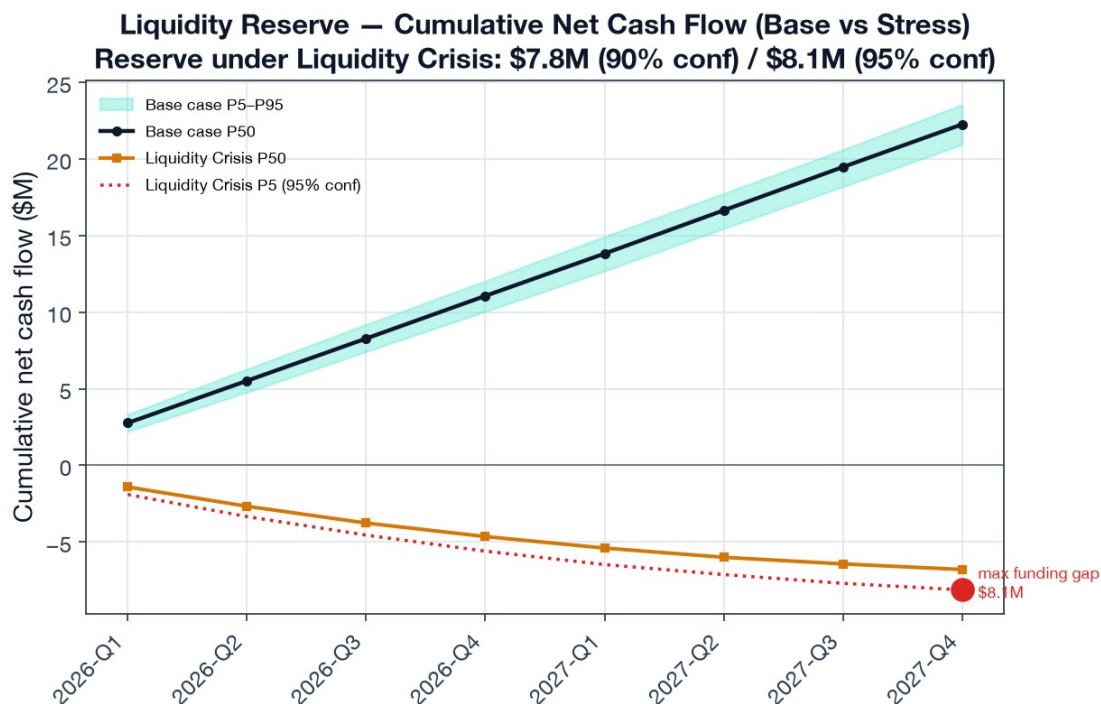
Stress scenarios

Transparency note. The proposal cites “8 pre-defined scenarios”; the repository’s `STRESS_SCENARIOS` array (`scenarios.ts:35-103`) currently defines 7 (base case + 6 stress scenarios). All seven are reported above. Adding an eighth (e.g. a stagflation / rate-shock scenario) is a one-line addition to that array.

3.4 Liquidity reserve at 90% / 95% confidence

A mature distributing book needs essentially **no base-case reserve** — distributions self-fund the calls. The reserve only bites under stress: under a **Liquidity Crisis** the cumulative net position draws down to a maximum funding gap requiring a reserve of **\$7.8M (90% confidence) / \$8.1M**

(95% confidence). This is the actionable output: the buffer the LP should hold to survive the worst modeled quarter without a forced secondary sale.



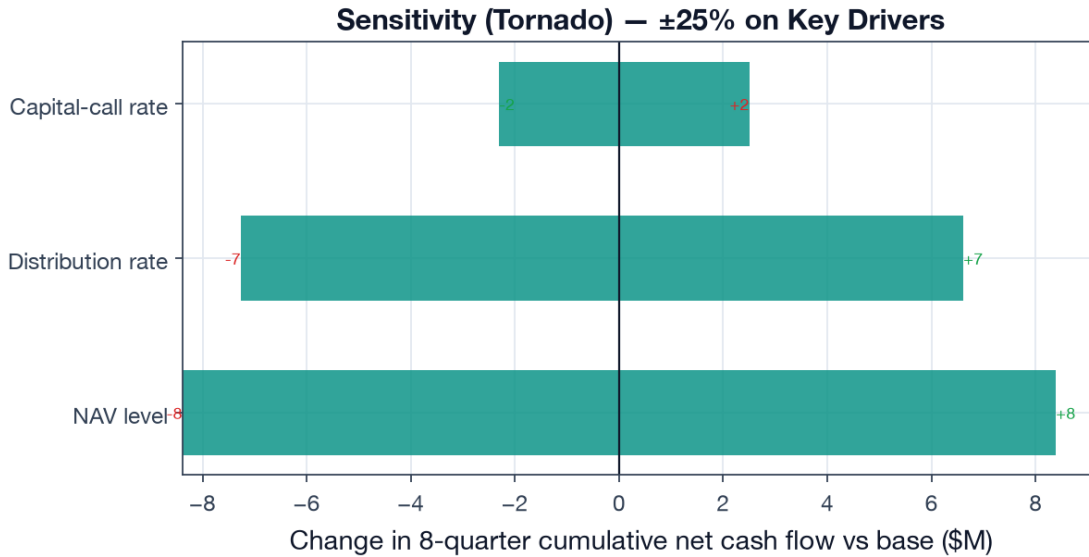
Liquidity reserve

3.5 Sensitivity — what actually moves the forecast ($\pm 25\%$)

A one-at-a-time $\pm 25\%$ perturbation of three key drivers ranks them by their swing on 8-quarter cumulative net cash flow:

Driver	Swing on cumulative net cash flow ($\pm 25\%$)	
NAV level	$\pm \\$16.8\text{M}$	
Distribution rate	$\pm \$13.9\text{M}$	
Capital-call rate	$\mp \$4.8\text{M}$	

NAV level and distribution rate dominate; the call rate matters $\sim 3\times$ less because the book is mostly deployed (only \$39M unfunded). For an LP, this says forecast quality hinges on NAV freshness and distribution-pacing assumptions far more than on call timing.



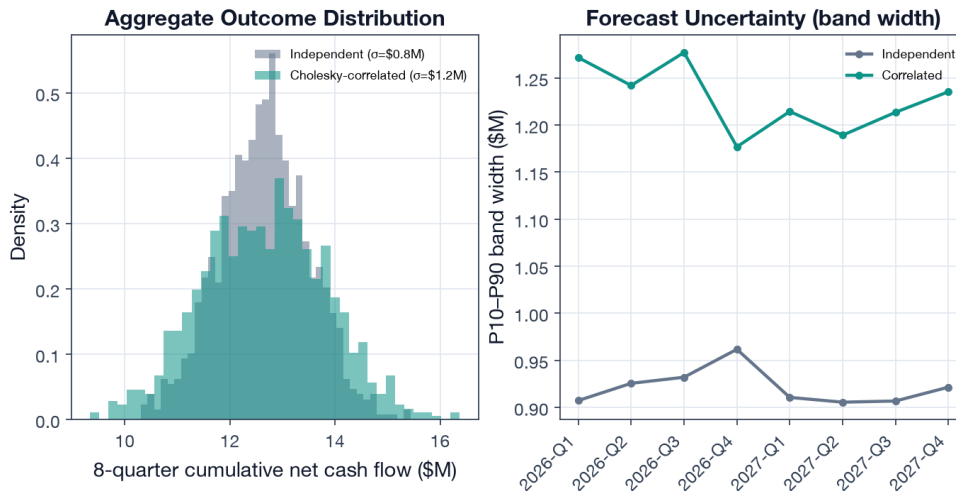
Sensitivity tornado

3.6 Cross-asset correlation — and an honest engine limitation

The Cholesky correlation model is a headline feature, but it surfaces a **real, documented limitation**: with 16 funds sharing only 6 asset classes, the per-fund correlation matrix has **duplicate rows** → **it is singular** → **Cholesky fails** → **the engine correctly falls back to independent simulation** (monte-carlo.ts:336-341). So on the full book, correlation does **not** engage.

To demonstrate the correlation path *does* work, we run a **5-fund, one-per-class** sub-portfolio where the 5×5 matrix is positive-definite. Turning correlation on widens the aggregate outcome distribution by **+39% in standard deviation** (σ \$1.17M vs \$0.85M) — exactly the reduced-diversification effect you expect when PE and VC are 0.55–0.80 correlated.

Cross-Asset Correlation Effect (5-fund one-per-class demo) — correlation reduces diversification, widening tails



Correlation effect

4. Findings

1. **The engine meets the forecast-accuracy target** — 5.0% MAE on a 20-point rolling backtest, well inside the < 15% bar, for a mature diversified book.
2. **Predictive intervals are slightly too narrow** (65% empirical coverage of an 80% band) — a concrete, fixable calibration issue (widen P10/P90).
3. **Reserve sizing is stress-driven, not base-case driven** — the product's value is in quantifying the ~\$8M buffer needed to survive a liquidity crisis, not the (≈ 0) base-case reserve.
4. **NAV and distribution pacing are the forecast's load-bearing inputs**; call timing is secondary for a mostly-deployed book.
5. **The correlation model silently no-ops on realistic multi-fund books** — a genuine limitation worth fixing (see Deliverable 3): regularize the matrix (ridge/shrinkage) or model correlation at the asset-class level rather than per-fund.

5. Limitations

- Synthetic-but-calibrated history, not live client cash flows (\$2). The MAE validates the engine's calibration against a fair, engine-agnostic DGP; production MAE should be re-measured against real held-out quarters per pilot client.
- Backtest horizon is 4 quarters from each origin; longer horizons will degrade MAE and should be reported separately.
- Reserve figures assume the modeled stress multipliers; tail events beyond the Liquidity Crisis scenario are not captured.

6. Reproducibility

from capstone/

python deliverable-2-forecasting-backtest/run_backtest.py

Outputs: results/forecast_metrics.json, six figures under figures/. Engine:
lib/onelp_eval/mc_engine.py (faithful port of src/lib/forecasting/). Portfolio +
history: lib/onelp_eval/portfolio.py.